

ALAN ORDORICA

Software Engineer | Backend Systems | Cloud & Platform

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Summary

Multipotentialite¹ software engineer with a passion for learning and building across disciplines—quickly picking up tools from backend systems to embedded hardware and turning ambiguous challenges into deliverable solutions

Experience

 **Google** Sunnyvale, CA
Software Engineer | Google Cloud May 2022 - Dec 2024

- Designed and implemented a high-fidelity control-plane simulator in C++/Python using gRPC and Protobuf, complete with pub/sub messaging and concurrency primitives (channels, callbacks). This simulator replicated complex Andromeda network behaviors, enabling developers to test new features in isolation and reducing full regression runtimes by **25%** (saving ~1.5 SWE-years each year).
- Developed a multicast VPC-gateway emulation layer by extending the simulator to support 1,000+ gateways connected to a single controller. Created new C++ modules and Python orchestration scripts to emit and process messages, mocked missing data-plane components (GSLB, ESF port), and helped customers validate scalable pub/sub topologies—reducing customer onboarding time by **30%**.
- Replaced brittle monitoring hooks with modern observability: migrated from BNS watchers to Streamz/PK-based monitors. Wrote Python scripts to query Monarch for job health, parse Borg metadata, and auto-restart failed services; integrated these into the test harness, cutting device downtime by **27%** and eliminating false alarms.
- Built debugging and orchestration tools for isolated test sandboxes. Provisioned per-test sandboxes in Borg/Sandman, scripted metrics collection and auto-recovery, and authored log-parsing utilities that quickly pinpointed broken runfile paths or output misconfigurations—reducing release-blocking test failures by **30%** and speeding developer feedback loops by **40%**.
- Created pre-population strategies for Spanner test databases. Developed Python utilities to seed test DBs via textproto, Spanner backups, and Sandman snapshots; configured cron jobs and multi-sandbox topologies to ensure reproducible datasets and cut environment setup time significantly.
- Contributed to host-agent re-architecture, replacing an OpenFlow-based on-host agent with a new C++ On-Host Controller (OHC), integrating control-plane logic directly on VM hosts. Updated gRPC interfaces and host-path plumbing, improving reliability and deprecating legacy code.

 **Apple** Sunnyvale, CA
Tools and Automation Engineer | CI Tools-Site Reliability July 2020 - March 2021

- Standardized deployment of automated test racks by planning host topology, cabling, and rack configuration and documenting bring up and handoff procedures to streamline operations.
- Engineered a reproducible provisioning pipeline in Python, Bash, and Ansible to automate firmware flashing, disk imaging, credential setup, and MDM enrollment—ensuring consistent, hands off device bring up.
- Built SSH-based orchestration and monitoring utilities with parallel scripting and Splunk integration to automate fleet health checks, alerting, and imaging workflows—enabling rapid remediation from the terminal.

 **Full Stack Developer** Remote
Software Engineer January 2020 - Present

- Developed modular React frontend and Node.js backend portfolio web apps with Docker Compose for local orchestration, enabling rapid environment setup.
- Developed Node.js RESTful microservices and ESP32-based automation projects using Python, Bash, and other tools, showcasing proficiency in hardware-software integration and system-level programming.

Technical Skills

Languages	Backend & API	Frontend & UI	DevOps & Cloud	Embedded Systems
- C++ - Python - Dart - SQL - Bash - JavaScript	- REST - gRPC - Concurrency - Networking - Protobuf	- React - Flutter	- GCP - Ansible - CI/CD - Git - Metrics - Docker - Logging - Monitoring - Kubernetes	- ESP32 - Linux - I²C - SPI - BLE - GDB

Education

San Francisco State University San Francisco, CA
Bachelor of Science Computer Engineering

¹ **Multipotentialite**: A person with multiple interests and an ability to learn quickly across different domains